



Lab 03

Bipolar Junction Transistor (BJT) Characteristics

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3.1 Objective

To investigate the characteristics of Bipolar Junction Transistor and its input and output characteristic curves in Common Emitter configuration with load line.

The CLOs, LDLs and PLOs mapped on in this lab are:

CLO	Description	LDL	PLO
CLO 1	Investigate characteristics of basic electronic devices and their application in development of electronic solutions.	Psy - 3	PLO-4
CLO 3	Apply appropriate techniques of OrCAD PSpice software for simulating electronic circuits.	Psy - 4	PLO-5
CLO 4	Demonstrate ethical values by complying with instructions and practicing punctuality and integrity	Aff - 3	PLO-8

Note: All the plots required to be attached should be labeled properly, followed by Task number and must be readable, i.e., use white background, suitable marker color (recommended – any dark color) and increase the width of trace. On PSpice there is an option to add text labels in the plots.



3.2 Pre-Lab Task

Lab 03

Bipolar Junction Transistor (BJT) Characteristics

Name: Syed Sualeh Abbas

Student ID: 10303

3.1 Objective

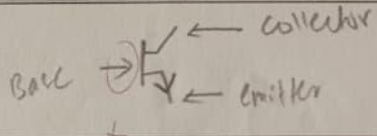
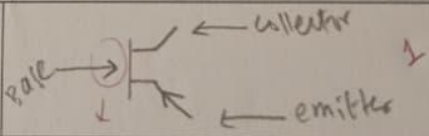
To investigate the characteristics of Bipolar Junction Transistor and its input and output characteristic curves in Common Emitter configuration with load line

3.2 Pre-Lab Task

Task 1: To draw the symbol of BJT and label its terminals and junctions

Name the three terminals and two junctions of BJT. Compare the terminals with respect to doping and thickness. Draw the symbols of *pnp* and *npn* BJTs with their terminals labeled and complete Table 3.1.

Table 3.1: BJT terminals and representation

Terminals of BJT		Doping	Thickness
1	Emitter	Heavily doped ✓	Moderate ✓
2	Base	Lightly doped ✓	Very thin ✓
3	collector	Moderately doped ✓	Thick ✓
Junction 1		Emitter-base (EB junction) ✓	
Junction 2		Base-collector (BC junction) ✓	
Symbolic Representation			
			
npn BJT		pnp BJT	

3.3 Introduction to Bipolar Junction Transistor

Bipolar transistors are made of either silicon (Si) or germanium (Ge). Their structure consists either of two layers of n-type material separated by a layer of p-type material (nnp), or of two layers of p-material separated by a layer of n-material (pnp). In either case, the center layer forms the base of the transistor, while the external layers form the collector and the emitter of the transistor. The symbolic representation must have been obtained in Pre-Lab Task. Note that the arrow at the emitter terminal of the transistor symbol for either type of transistor points in the direction of conventional current flow and thus provides a useful reference. One part of this experiment will demonstrate how you can determine the type of transistor and its material, and identify its three terminals.

The relationships between the voltages and the currents associated with a bipolar junction transistor under various operating conditions determine its performance. These relationships are collectively known as the characteristics of the transistor which will be investigated in later part.

3.4 Modes of Operation

There are 3 modes of operation of a BJT as given in Table which are categorized by the biasing of the two junctions; emitter-base (EB) junction and collector-base (CB) junction.

Table 3.4 a: Modes of operation of BJT

Modes of Operation	Base-Emitter Junction	Base-Collector Junction
Cut-off Region	Reverse-Biased	Reverse-Biased
Saturation Region	Forward-Biased	Forward-Biased
Linear / Active Region	Forward-Biased	Reverse-Biased

For *nnp* configuration the conditions for base emitter junction and base-collector Junction Voltages in Reverse and Forward bias mode would be as follows:

Table 3.5b: Modes of operation of BJT

Modes of Operation	Base-Emitter Junction		Base-Collector Junction	
Cut-off Region	Reverse-Biased	$V_E > V_B$	Reverse-Biased	$V_C > V_B$
Saturation Region	Forward-Biased	$V_B > V_E$	Forward-Biased	$V_B > V_C$
Linear / Active Region	Forward-Biased	$V_B > V_E$	Reverse-Biased	$V_C > V_B$

For *pnp* configuration the conditions for base-emitter junction Voltage and base-collector Junction Voltage in Reverse and Forward bias mode would be as follows:

Table 3.6 c: Modes of operation of BJT

Modes of Operation	Base-Emitter Junction		Base-Collector Junction	
	Reverse-Biased	$V_B > V_E$	Reverse-Biased	$V_B > V_C$
Cut-off Region	Reverse-Biased	$V_B > V_E$	Reverse-Biased	$V_B > V_C$
Saturation Region	Forward-Biased	$V_E > V_B$	Forward-Biased	$V_C > V_B$
Linear / Active Region	Forward-Biased	$V_E > V_B$	Reverse-Biased	$V_B > V_C$

To investigate the operation of BJT in these regions and relation between voltage and current at its input and output terminals, we will use the most frequently used transistor configuration called common-emitter configuration as the emitter is common to both the input and output terminals (in this case common to both the base and collector terminals) as shown in Figure 3.2. Two sets of characteristics are necessary to describe fully the behavior of the common-emitter configuration: one for the input or base-emitter circuit and one for the output or collector-emitter circuit.

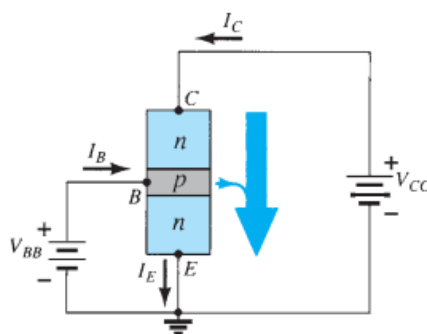


Figure 1: Common-emitter configuration

These characteristics are represented by obtaining VI curves i.e., base-current against base-emitter voltage (I_B - V_{BE} curve) and collector current against collector-emitter voltage (I_C - V_{CE} curve). When base-emitter junction gets forward biased, at a given base-emitter voltage a constant base-current is established. Output characteristics also called common-emitter characteristics i.e., plot of collector current against collector-emitter voltage, can represent the operation of BJT in different modes.

At small values of V_{CE} , the collector voltage is less than that of base causing CB junction to get forward biased. This causes the transistor to enter saturation region where both the junctions are forward biased. For a given base bias, increase in V_{CE} reduces the forward bias and eventually reverse bias the CB junction causing it to operate in active (linear) region. In this region, the collector current ideally should not increase once the CB junction becomes reverse biased. But practically due to early effect, the width of depletion region is affected by reverse biased voltage which increases the reverse saturation current. This makes the collector current to slightly increase at higher values of V_{CE} causing the characteristics to exhibit some slope.

Task 2: To investigate the input characteristics of 2N3904 npn-BJT using PSpice

Figure 3.3 shows 2N3904 npn transistor connected in common emitter configuration. Its input characteristics can be investigated through plot between base current I_B and base-emitter voltage V_{BE} at a given collector-emitter voltage.

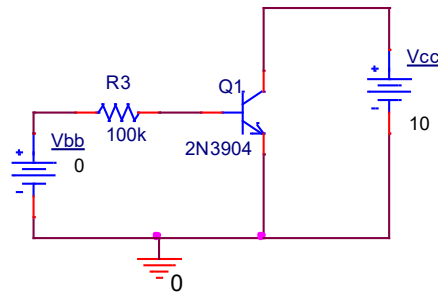
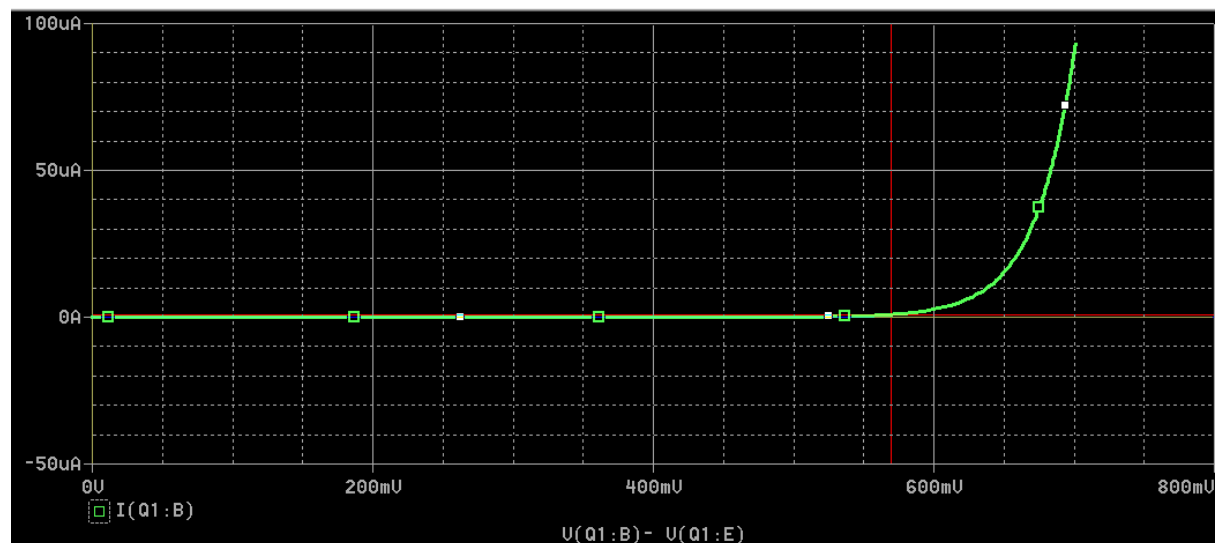


Figure 3.3: BJT 2N3904 connected in common-emitter configuration

1. Implement the circuit using OrCAD Capture. Set V_{cc} to 10 V which indicates that collector-emitter voltage is also set to 10 V as the source is directly connected to collector terminal.
2. To obtain curve between base current and base-emitter voltage, the base current will be varied by varying the voltage of source V_{bb} . To do this, perform DC sweep analysis taking V_{bb} as sweep variable. Vary its voltage from 0 to 10 V at an increment of 0.1V.
3. Run the simulation and obtain trace for base current. Set the x-axis variable to base-emitter voltage. Go to Axis Settings, X-Axis tab and change x-axis variable through the required expression e.g. $V(Q1:B)-V(Q1:E)$.
4. Attach the plot with suitable title and white background color. Analyze the input characteristics through obtained plots. Can you relate the plot with VI-characteristics of diode?

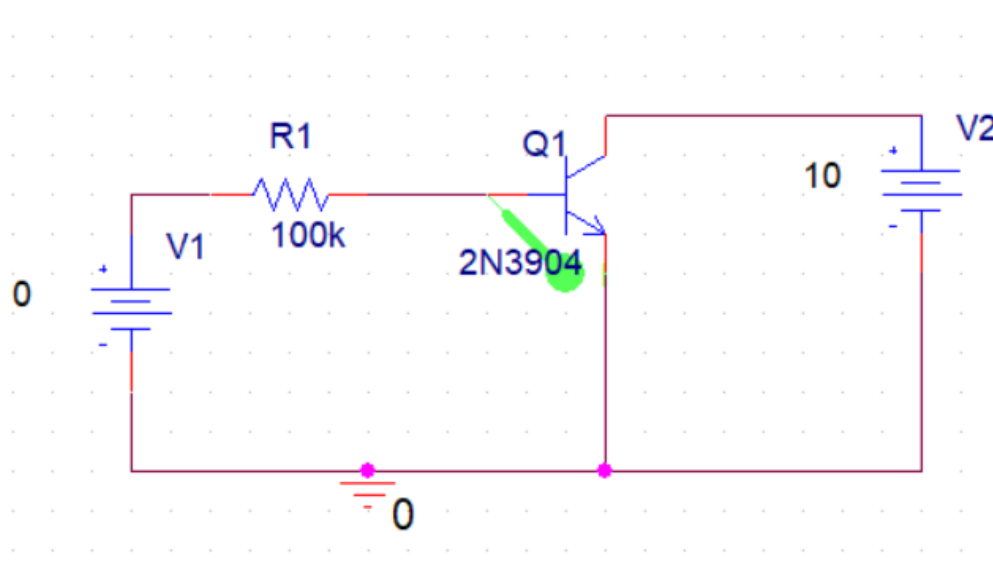
RELATION BETWEEN BASE-EMITTER VOLTAGE & BASE CURRENT



The obtained plot shows the variation of base current (I_B) with respect to base emitter voltage (V_{BE}) while keeping collector emitter voltage ($V_{CE} = 10$ V) constant. This represents the input characteristic of a BJT in common-emitter configuration.

From the graph, it is observed that for low values of V_{BE} , the base current remains almost zero. This indicates that the base emitter junction is not forward biased and the transistor is in the cut-off region. As V_{BE} increases beyond approximately 0.6 V, the base current starts increasing rapidly and nonlinearly. A small increase in V_{BE} results in a large increase in I_B , showing an exponential relationship.

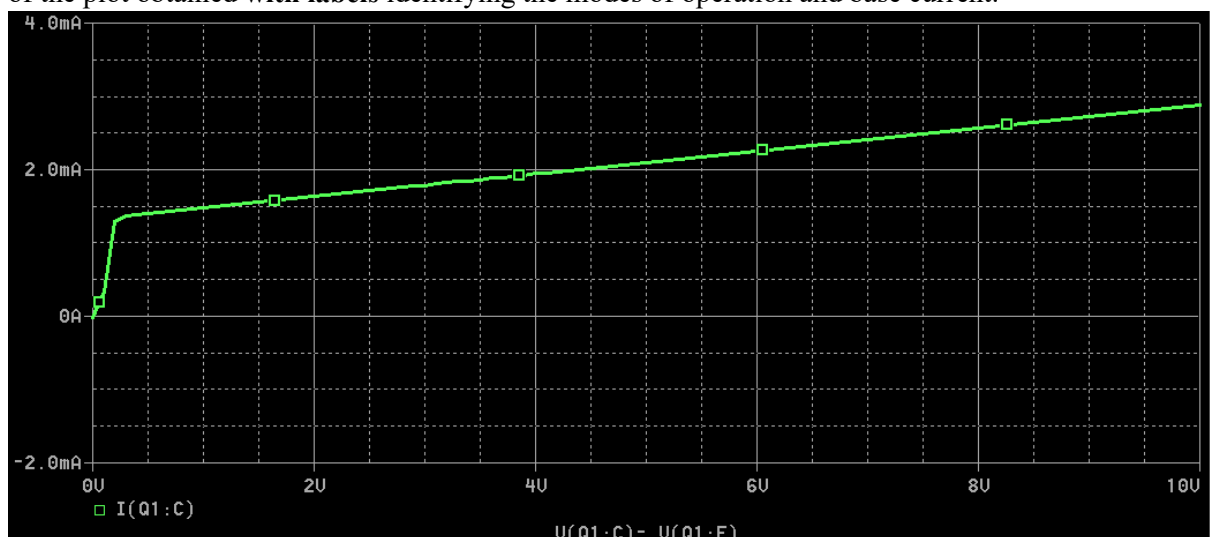
This behavior is very similar to the V–I characteristics of a diode. The reason is that the base emitter junction of a BJT is a PN junction, just like a diode. When forward biased, it follows the diode equation, leading to an exponential rise in current after the cut-in voltage. small deviations from a diode curve are due to transistor specific effects such as base resistance.



Task 3: To investigate the output characteristics of 2N3904 npn-BJT using PSpice

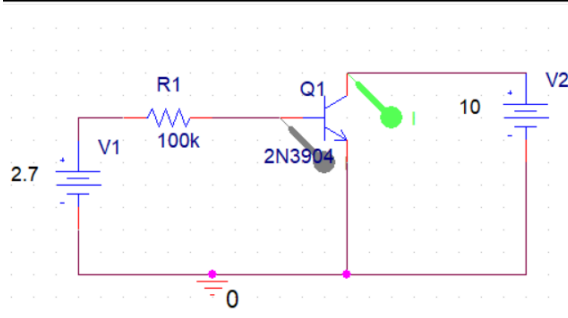
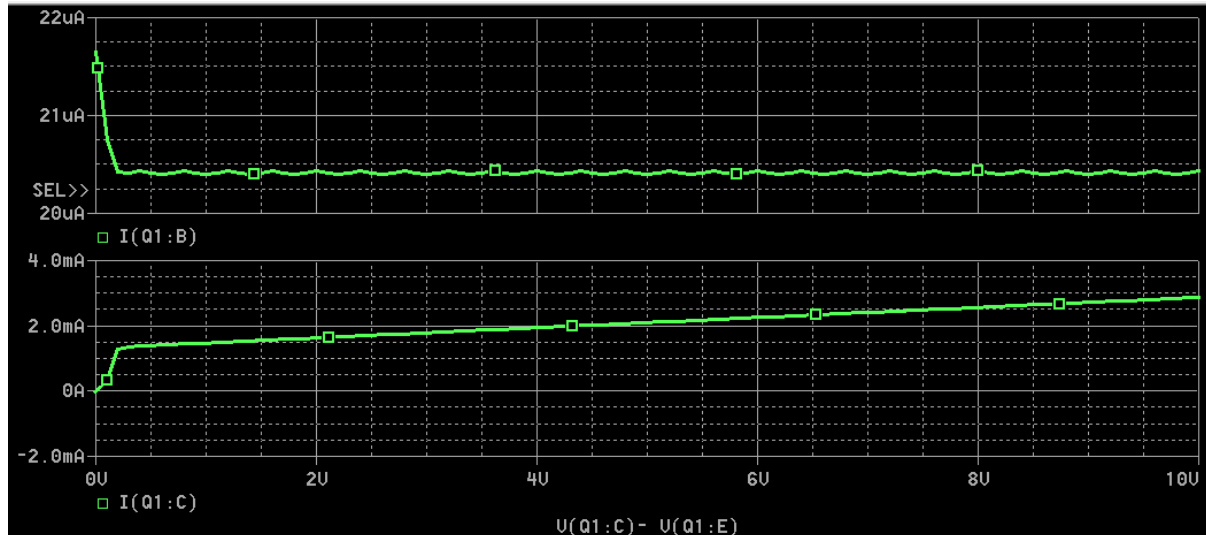
To investigate the output characteristics of BJT, collector current is plotted against collector-emitter voltage at different values of base current.

1. Modify the circuit implemented in previous task. Initially set V_{bb} to 2.7 V. Create a new simulation profile and perform DC sweep analysis to vary collector to emitter voltage by selecting V_{cc} as primary sweep variable. Vary this voltage from 0 to 10V at an increment of 0.1V. Run the simulation to obtain trace for collector current I_C and make sure the x-axis variable is set to V_{CE} (e.g. $V(Q1:C)-V(Q1:E)$).
2. From I_C - V_{CE} curve, identify the modes of operation of BJT and add suitable labels. **Attach** graph of the plot obtained **with labels** identifying the modes of operation and base current.



3. Measure the base current also for which this plot is obtained. You can do this by adding trace for I_B in a separate window. **Attach graph of the plot obtained with labels.**

BASE CURRENT (1st Graph)



1. Saturation Region

At low values of V_{CE} (approximately 0–0.2 V), the collector current rises sharply with a small increase in V_{CE} . In this region, both the base–emitter and base–collector junctions are forward biased. The transistor is said to be in saturation, and it cannot maintain a constant collector current. This region is used when the transistor operates as a switch (ON state).

2. Active Region

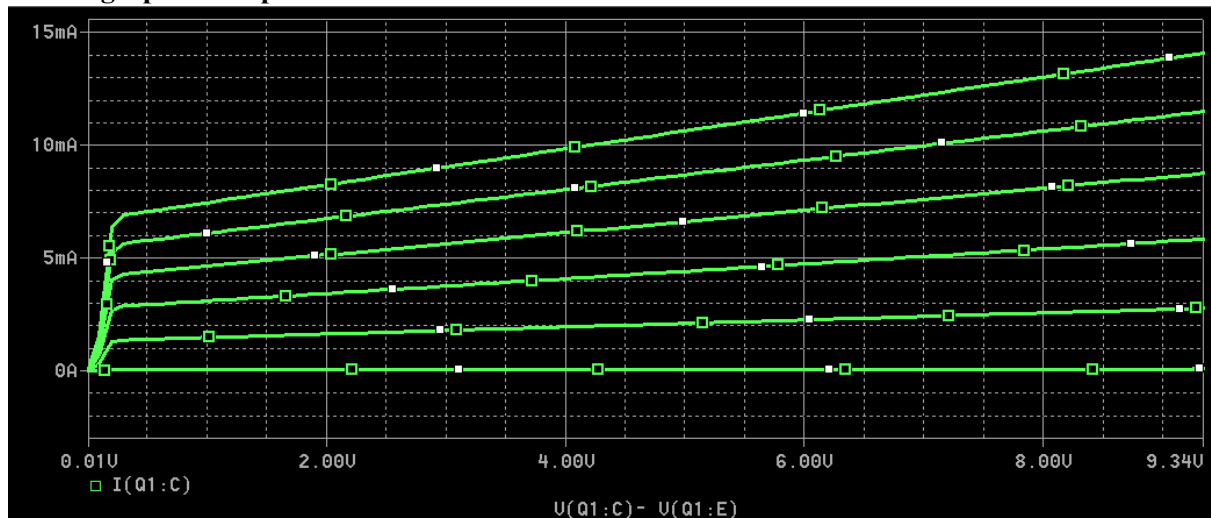
As V_{CE} increases beyond the saturation voltage, the transistor enters the active region. In this region, I_C remains almost constant despite an increase in V_{CE} , as seen by the nearly horizontal portion of the curve. Here, the base–emitter junction is forward biased while the base–collector junction is reverse biased. The collector current is mainly controlled by the base current (set by V_{bb}), making this region suitable for amplification.

3. Cut-off Region

Although not clearly visible in this plot (because V_{bb} is sufficiently high), the cut-off region corresponds to very small or zero collector current when the base current is nearly zero. In this region, both junctions are reverse biased and the transistor is in the OFF state.

4. Now, to investigate the effect of base current on this relation, the same primary sweep will be performed for different values of base current. To do this, edit simulation settings and check the secondary sweep option. Take V_{bb} as secondary sweep variable with its value ranging from 0.7V to 10.7V at an increment of 2V.

5. Run the simulation and obtain I_C - V_{CE} curves. With each curve **label** the corresponding base current. **Attach graph of the plot obtained with labels.**



6. What is effect of increasing base current on the obtained collector current curves?

Effect of Increasing Base Current:

As the base current increases, the collector current (I_C) increases proportionally for a given collector-emitter voltage (V_{CE}). This is evident from the family of curves, where each successive curve (moving upward) represents a higher base current level. The curves show that the transistor operates in the active region, where I_C is primarily controlled by the base current, demonstrating the current amplification property of the BJT. The relatively flat nature of the curves in the active region indicates that collector current remains nearly constant for varying V_{CE} at a fixed base current, with the spacing between curves representing the current gain (β) of the transistor.

7. Analyze the relation between collector current with collector to emitter voltage in each mode of operation?

Analysis of I_C vs V_{CE} Relationship in Different Operating Modes:

The collector current exhibits distinct behavior across different operating regions. In the cut-off region, represented by the bottom curve where base current is approximately zero, the collector current remains nearly zero regardless of the collector-emitter voltage, indicating the transistor is OFF.

In the saturation region, occurring at low V_{CE} values (typically below 0.5V), the collector current rises steeply as both junctions become forward biased, and the transistor behaves like a closed switch.

As V_{CE} increases beyond the saturation region, the transistor enters the active region where the curves flatten out and become nearly horizontal. In this region, the collector current remains relatively constant for a given base current despite variations in V_{CE} , demonstrating that I_C is primarily controlled by the base current. This flat characteristic in the active region is essential for amplification applications, as it allows the transistor to maintain stable current gain across varying load conditions.

8. Does collector current remain constant in active region? What would have been the ideal behavior?

Collector Current Behavior in Active Region:

No, the collector current does not remain perfectly constant in the active region. As observed in the characteristic curves, there is a slight upward slope in each curve as V_{CE} increases, indicating a gradual increase in collector current with increasing collector-emitter voltage. This non-ideal behavior is primarily due to the Early effect (base-width modulation), where increasing V_{CE} causes the collector-base depletion region to widen, effectively reducing the base width and increasing the current gain.

Ideal Behavior:

Ideally, the collector current should remain perfectly constant in the active region for a fixed base current, resulting in completely horizontal characteristic curves. In an ideal transistor, I_C would depend only on the base current ($I_C = \beta \times I_B$) and would be independent of V_{CE} . This would yield infinite output resistance, making the transistor a perfect current source controlled solely by the base current. However, real transistors exhibit finite output resistance due to the Early effect, resulting in the slight positive slope observed in practical characteristic curves.

3.5 DC Current Gain

In DC mode, the levels of I_C and I_B for any transistor are related by a quantity called beta and defined

by the following equation: $\beta = \frac{I_C}{I_B}$

Where, I_C and I_B are determined at a particular operating point on the characteristics. For practical devices the level of β typically ranges from about 50 to over 400. This gain reveals the relative magnitude of collector current with respect to the base current. On specification sheet, it is usually included as h_{FE} .

Task 4: To analyze variation in DC current gains with collector-emitter voltage and base current

- From the plots obtained in previous task, through cursors measure the collector current at two given points of each curve; when V_{CE} is 2V and when it is 10V. Record your measurements in Table 3.7 and mention corresponding base current.

Table 3.7: Data collection for DC current gain and its calculated values

Base Current: I_B (μA)	Collector-Emitter Voltage: $V_{CE}=2V$		Collector-Emitter Voltage: $V_{CE}=10V$	
	I_C (mA)	β	I_C (mA)	β
99.98	8.25	82.51	14.616	146.18
80.03	6.744	84.26	11.931	149.08
60.11	5.139	85.49	9.0778	151.02
40.22	3.428	85.23	6.049	150.39
20.43	1.633	79.93	2.877	140.82



2. Calculate the DC gain β for each. Is there any variation in DC current gain as we move from left to right on the curve?

Yes, there is clear variation in the DC current gain. The β values increase substantially as V_{ce} increases from 5V to 10V, with an average increase of approximately 70-75%. This variation is primarily attributed to the Early effect, where higher collector-emitter voltages cause base-width modulation, resulting in increased current gain. Additionally, there is slight variation in β at different base current levels within the same V_{ce} , though this is less pronounced. This non-ideal behavior demonstrates that β is not strictly constant but depends on the operating point of the transistor, particularly the collector-emitter voltage.

3. This current gain β is the property of transistor and does vary by temperature as well. Go through the datasheet of 2N3904 BJT (<https://www.onsemi.com/pub/Collateral/2N3903-D.PDF>) and observe how this parameter is specified in the datasheet. The AC current gain is specified as h_{fe} in the datasheet (with small letters in subscript) which is something different than the DC gain. The curve shown in Figure 15 of datasheet illustrates the effect of temperature on h_{FE} . Analyze it briefly.

The DC current gain (h_{FE}) of the 2N3904 transistor depends on the collector current (I_C), collector-emitter voltage (V_{CE}), and temperature. As temperature rises, h_{FE} increases due to enhanced carrier injection. However, h_{FE} decreases at very high or very low collector currents. The Early effect slightly increases h_{FE} at higher V_{CE} values. Therefore, transistor circuits should be designed while considering temperature variations, as h_{FE} is not a constant parameter.

Task 5: To investigate the change in operating point with the help of load line

The collector current together with collector to emitter voltage (I_C , V_{CE}) specifies operating point of transistor. The operating point is a fixed point on the characteristics; it is also called the quiescent point (Q -point). It is interesting to note that because the base current is controlled by the level of R_B and I_C is related to I_B by a constant β , the magnitude of I_C is not a function of the resistance R_C . Changing R_C to any level will not affect the level of I_B or I_C as long as we remain in the active region of the device. However, as we shall see, the level of R_C will determine the magnitude of V_{CE} , which is an important parameter.

1. To verify this, connect a $1k\ \Omega$ resistor at collector terminal of transistor. Set V_{bb} of 2.7V and V_{CC} at 10V. Simulate the circuit and obtain collector-to-emitter voltage and collector current and note down in Table.

```

NODE   VOLTAGE   NODE   VOLTAGE   NODE   VOLTAGE   NODE   VOLTAGE
(N00570)  2.7000 (N00578)  .6584 (N00739)  10.0000 (N00743)  7.5102

VOLTAGE SOURCE CURRENTS
NAME      CURRENT
V_V1      -2.042E-05
V_V2      -2.490E-03

TOTAL POWER DISSIPATION  2.50E-02  WATTS

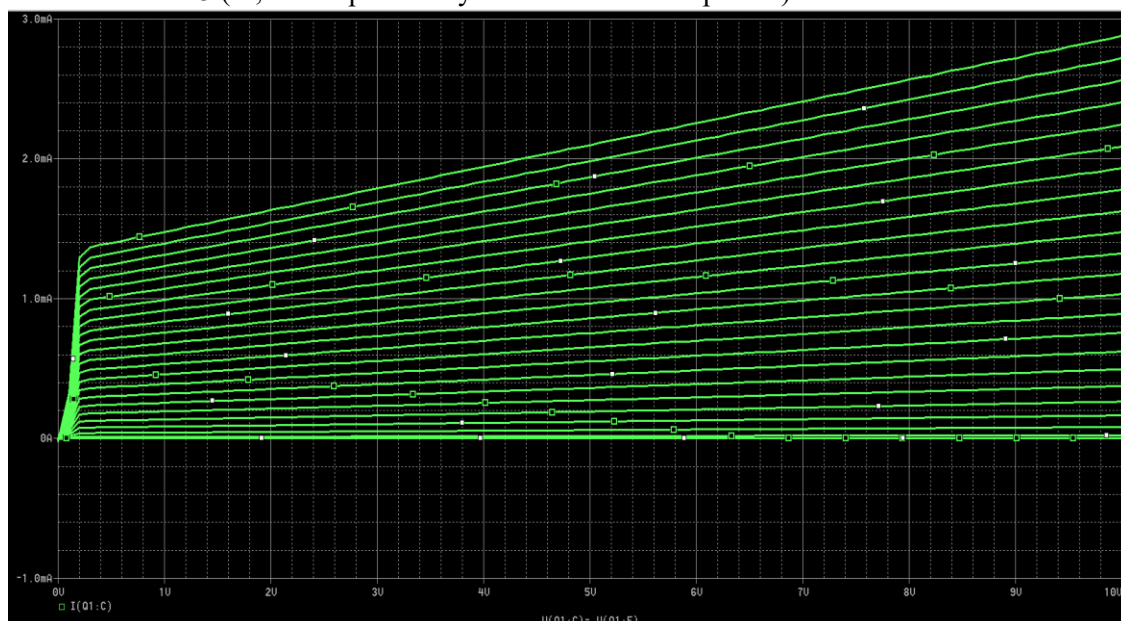
JOB CONCLUDED
**** 02/10/26 00:00:13 ***** PSpice Lite (March 2016) ***** ID# 10813 ****
** Profile: "SCHEMATIC1-sim1" [ C:\Users\Administrator\OneDrive - Habib University\Sem 4\Basic Electronics\Lab
**** JOB STATISTICS SUMMARY
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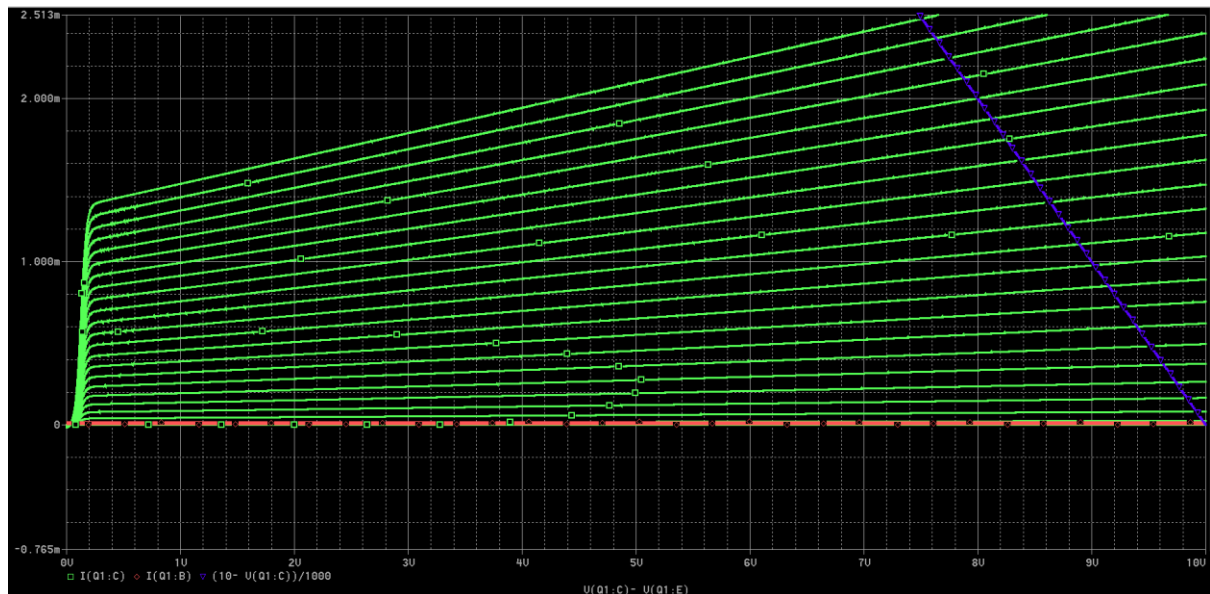
Table 3.8: Operating point of BJT with 1k Ohm load resistor

Collector-Emitter Voltage: V_{CE}	7.51V
Collector Current: I_C	2.49mA

- To directly obtain the operating point of transistor for a given load resistor and supply voltage V_{CC} , a load line can be plotted over the characteristic curves. The I_C - V_{CE} characteristics curves obtained in Task 3 for different values of base current can be used to find operating point by adding load line. Remove the $1k\Omega$ from circuit and through Primary and Secondary Sweep obtain I_C - V_{CE} curves as done in Task 3 (or, re-use previously created simulation profile).



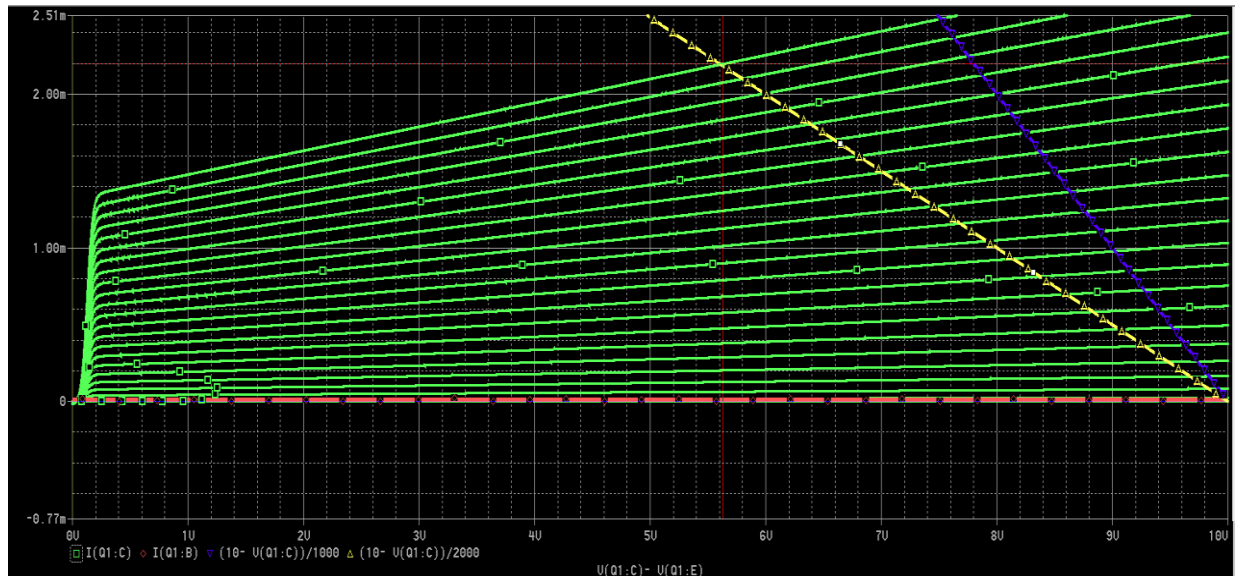
3. To add a load line for $1k\Omega$ load resistor R_C , add a trace with expression $(10-V(Q1:C))/1000$, which is PSpice trace expression for $I_C = \frac{(V_{CC} - V_{CE})}{R_C}$ at fixed $10V$ V_{CC} and $1k\Omega$ load resistor R_C when V_{CE} is being varied due to change in I_C as base current is varied. The intersection of the obtained load line with each characteristic curve gives the operating point of transistor for the corresponding base current. Note that instead of V_{CE} , collector voltage is used in PSpice trace expression as emitter is directly grounded.
4. You can verify that the operating point found in Table 3.8 is the intersection point of load line and the I_C - V_{CE} curve at base current of $20\mu A$. Attach the obtained plot and state your observations in the box below:



Trace Color	Trace Name	Y1	Y2	Y1 - Y2	Y1(Cursor1) - Y2(Cursor2)	Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)	Max Y	Min Y	Avg Y
	X Values	7.5145	0.000	7.5145	Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)				
CURSOR 1,2	I(Q1:C)	2.4898m	-23.65E-21	2.4898m	0.000	0.000	2.4898m	-23.65E-21	1.2449m	
	I(Q1:B)	20.417u	-10.208f	20.417u	-2.4694m	-10.208f	20.417u	-10.208f	10.208u	
	(10- V(Q1:C))/1000	2.4860m	9.999m	-7.5130m	-3.7959u	9.999m	9.999m	2.4860m	6.2425m	

by examining the intersection points between the load line and the characteristic curves, we can verify the operating point. Based on the DC bias point analysis from Table 3.8 ($I_C = 2.49mA$, $V_{CE} = 7.51V$), the corresponding Q-point should lie on one of the middle characteristic curves at $I_B \approx 20\mu A$.

5. To find the effect of varying load resistor, add another trace of load line with R_c of $2k\Omega$. Attach the capture of obtained plots with two load lines; one with $1k\Omega$ and other with $2k\Omega$ resistor. State your observations in the box below:



Trace Color	Trace Name	Y1	Y2	Y1 - Y2
	X Values	5.6310	0.000	5.6310
CURSOR 1,2	I(Q1:C)	2.1973m	-23.65E-21	2.1973m
	I(Q1:B)	20.417u	-10.208f	20.417u
	(10- V(Q1:C))/1000	4.3700m	9.999m	-5.6290m
	(10- V(Q1:C))/2000	2.1850m	5.0023m	-2.8173m

Load Line Comparison:

1kΩ Load Line (Blue):

- Extends from saturation point ($V_{CE} = 0V$, $I_C = V_{CC}/R_C = 10V/1k\Omega = 10mA$) to cutoff point ($V_{CE} = 10V$, $I_C = 0mA$)
- Steeper slope: $-1/R_C = -1/1000 = -0.001$
- Higher maximum collector current capability

2kΩ Load Line (Yellow):

- Extends from saturation point ($V_{CE} = 0V$, $I_C = V_{CC}/R_C = 10V/2k\Omega = 5mA$) to cutoff point ($V_{CE} = 10V$, $I_C = 0mA$)
- Shallower slope: $-1/R_C = -1/2000 = -0.0005$
- Lower maximum collector current capability
- Both lines pass through the common point ($V_{CC}, 0$) = (10V, 0mA)

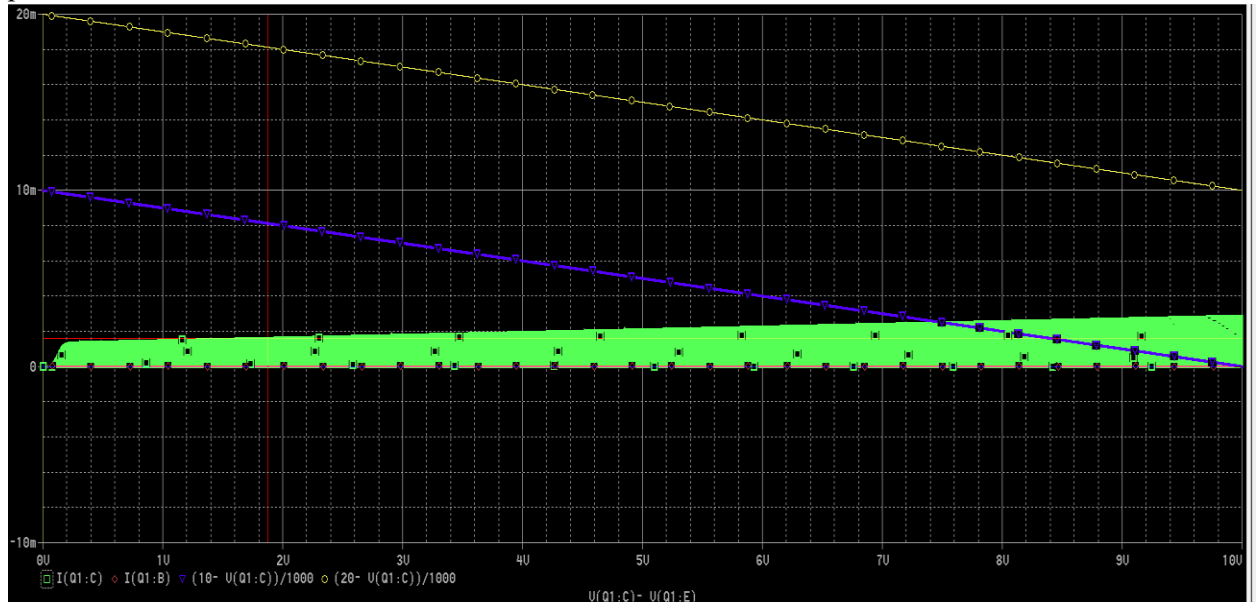
Effect of Increasing Load Resistor (R_C):

From the cursor data, at $V_{CE} = 5.63V$:

- 1kΩ load line: $I_C \approx 2.48mA$
- 2kΩ load line: $I_C \approx 2.19mA$

Varying the load resistor R_C changes the operating point position along a given characteristic curve but does not change the transistor's intrinsic behavior. The load resistor determines the circuit constraint (load line), while the base current determines which characteristic curve the transistor operates on.

6. Now, to find the impact of varying supply V_{cc} , remove the load line for $2k\Omega$ and add another load line with trace expression having $20V V_{cc}$. Attach the capture of plot.
7. Analyze the impact of base current I_B , supply voltage V_{cc} and load resistor R_c on the operating point of transistor.



Effects of Increasing I_B :

- Q-point moves upward and leftward along the fixed load line
- Collector current (I_C) increases linearly ($I_C = \beta \times I_B$, where β is transistor gain)
- Collector-emitter voltage (V_{CE}) decreases due to larger voltage drop across R_C
- Relationship: $V_{CE} = V_{CC} - (I_C \times R_C)$
- Transistor transitions from cutoff region through active region toward saturation

Parallel Shift:

- Both load lines have identical slopes since R_C is the same ($1k\Omega$)
- The $20V$ load line is a parallel outward translation of the $10V$ line
- Both lines would intersect if extended backward, meeting at the origin conceptually

Effect on Operating Range:

- The $20V$ supply doubles the maximum collector current capability (from $10mA$ to $20mA$)
- The cutoff voltage doubles from $10V$ to $20V$
- The active region is significantly expanded, providing much greater operational space

Effect on Q-Point (for same I_B):

- For any given base current (same characteristic curve):
 - Collector current remains approximately the same (minimal change)
 - V_{CE} increases substantially (approximately doubles)
 - Q-point shifts horizontally to the right along the characteristic curve



Assessment Rubric
Lab 03
Bipolar Junction Transistor (BJT) Characteristics

Name:	Student ID:
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Points Distribution

Task No.	LR2 Simulation	LR 4 Data Collection	LR5 Plots	LR 6 Calculation	LR 10 Analysis
Task 1	-	-	/8	-	-
Task 2	/4	-	/4	-	/4
Task 3	/8	-	/8	-	/8
Task 4	-	/8	-	/8	/4
Task 5	/8	/4	/8	-	/8
	/20	/12	/28	/8	/24
CLO Mapped	CLO 1	CLO 1	CLO 1	CLO 1	CLO 1

Affective Domain Rubric		Points	CLO Mapped
AR 3	Pre-Lab Submission	4	CLO 1
AR 4	Report Submission	4	

CLO-(LDL)	Total Points	Points Obtained
CLO1 – (Psy-3)	100	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.